

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250269220

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

Ramsay; Landon

SLIDABLE FREE-ROTATING BARBELL GRIP

Abstract

A rigid or semi-rigid grip adapted to slide and/or rotate freely on a barbell, operable to allow a user to activate stabilizer muscles and tendons and providing a free range of motion across during select weightlifting exercises, such as bench press, contributing to core strength training and core strength development.

Inventors: Ramsay; Landon (Salt Lake City, UT)

Applicant: Ramsay; Landon (Salt Lake City, UT)

Family ID: 1000007698633

Appl. No.: 18/585003

Filed: February 22, 2024

Publication Classification

Int. Cl.: A63B21/072 (20060101)

U.S. Cl.:

CPC A63B21/0724 (20130101);

Background/Summary

FIELD OF THE INVENTION

[0001] This invention relates to exercise equipment; and more particularly relates to grips for barbells.

BACKGROUND

Description of the Related Art

[0002] General fitness is an increasingly popular goal of modern populations. An aspect of general fitness is weightlifting in which barbells are used in lifts such as the bench press. In performing a bench press, shoulder press, curls, and other exercises, the palms are traditionally positioned statically on the barbell as the weights are moved. Although most exercises, including bench press, do not completely isolate any particular muscle group, such as the pectoral or triceps, the static positioning of the hands does prevent stabilizer muscles and tendons from being activated which would traditionally be necessary to stabilize a weight through a free range of motion. Additionally, the static positioning of the hands strains the wrists as the forearms move axially during lifts such as the bench press.

[0003] There exists no efficient methods, systems or apparatus in the art which allow for free range of motion of the hands while performing exercises using dumbbells and barbells. There is a need in the art for a simple and efficient means of converting a static grip on a barbell to free-moving grip.

SUMMARY

[0004] From the foregoing discussion, it should be apparent that a need exists for a free-moving barbell grip. The present invention has been developed in response to the present state of the art, and in particular, in response to the problems and needs in the art that have not yet been fully solved by currently available nesting solutions. Accordingly, the present invention has been developed to provide a free-rotating, detachable grip for a barbell, the grip comprising: an interrupted, semi-rigid sleeve dimensioned to have an interior circumference exceeding an exterior circumference of a barbell; wherein the sleeve comprises an exterior surface, a forward surface, and an interior surface, the sleeve defining a hollow interior recess adapted to receive a barbell; wherein the sleeve defines a slit operable to allow the grip to detachably affix over a barbell.

[0005] The grip may comprise polymeric materials. The interior surface may comprise texturing. The interior surface may comprise a plurality of rollers. The interior surface may comprise a plurality of ball bearings in some embodiments.

[0006] The grip may be operable to slide laterally on a barbell when grip pressure is released. The grip may be operable to slide axially on a barbell when grip pressure is released.

[0007] A second free-rotating, detachable grip for a barbell is provided, the grip comprising: an interrupted, semi-rigid sleeve dimensioned to have an interior circumference exceeding an exterior circumference of a barbell; wherein the sleeve comprises an exterior surface, a forward surface, and an interior surface, the sleeve defining a hollow interior recess adapted to receive a barbell; wherein the sleeve defines a slit operable to allow the grip to detachably affix over a barbell; wherein the grip is operable to slide laterally and axially on a barbell when grip pressure is released; wherein the grip is operable to lock in place forming a friction fit with the barbell when grip pressure is increased.

[0008] Reference throughout this specification to features, advantages, or similar language does not imply that all of the features and advantages that may be realized with the present invention should be or are in any single embodiment of the invention. Rather, language referring to the features and advantages is understood to mean that a specific feature, advantage, or characteristic described in connection with an embodiment is included in at least one embodiment of the present invention. Thus, discussion of the features and advantages, and similar language, throughout this specification may, but do not necessarily, refer to the same embodiment.

[0009] Furthermore, the described features, advantages, and characteristics of the invention may be combined in any suitable manner in one or more embodiments. One skilled in the relevant art will recognize that the invention may be practiced without one or more of the specific features or advantages of a particular embodiment. In other instances, additional features and advantages may be recognized in certain embodiments that may not be present in all embodiments of the invention.

[0010] These features and advantages of the present invention will become more fully apparent from the following description and appended claims, or may be learned by the practice of the invention as set forth hereinafter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] In order that the advantages of the invention will be readily understood, a more particular description of the invention briefly described above will be rendered by reference to specific embodiments that are illustrated in the appended drawings. Understanding that these drawings depict only typical embodiments of the invention and are not therefore to be considered to be limiting of its scope, the invention will be described and explained with additional specificity and detail through the use of the accompanying drawings, in which:

[0012] FIG. 1 illustrates a forward perspective of a free-rotating grip according to an embodiment of the present invention;

[0013] FIG. 2 illustrates a forward perspective of a free-rotating grip according to an embodiment of the present invention;

[0014] FIG. 3 illustrates an environmental perspective of free-rotating grips according to an embodiment of the present invention;

[0015] FIG. 4 illustrates a forward perspective of a free-rotating grip according to an embodiment of the present invention;

[0016] FIG. 5 illustrates a forward perspective of a free-rotating grip according to an embodiment of the present invention;

[0017] FIG. 6 illustrates a forward perspective of a free-rotating grip according to an embodiment of the present invention;

[0018] FIG. 7 illustrates a forward perspective of a free-rotating grip according to an embodiment of the present invention; and

[0019] FIG. 8 illustrates a forward perspective of a free-rotating grip and pry bar according to an embodiment of the present invention.

DETAILED DESCRIPTION

[0020] Reference throughout this specification to “one embodiment,” “an embodiment,” or similar language means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present invention. Thus, appearances of the phrases “in one embodiment,” “in an embodiment,” and similar language throughout this specification may, but do not necessarily, all refer to the same embodiment.

[0021] Furthermore, the described features, structures, or characteristics of the invention may be combined in any suitable manner in one or more embodiments. In the following description, numerous specific details are provided, such as examples of programming, software modules, user selections, network transactions, database queries, database structures, hardware modules, hardware circuits, hardware chips, etc., to provide a thorough understanding of embodiments of the invention. One skilled in the relevant art will recognize, however, that the invention may be practiced without one or more of the specific details, or with other methods, components, materials, and so forth. In other instances, well-known structures, materials, or operations are not shown or described in detail to avoid obscuring aspects of the invention.

[0022] The phrases “at least one”, “one or more”, and “and/or” are open-ended expressions that are both conjunctive and disjunctive in operation. For example, each of the expressions “at least one of A, B and C”, “at least one of A, B, or C”, “one or more of A, B, and C”, “one or more of A, B, or C” and “A, B, and/or C” means A alone, B alone, C alone, A and B together, A and C together, B and C together, or A, B and C together.

[0023] The term “a” or “an” entity refers to one or more of that entity. As such, the terms “a” (or “an”), “one or more” and “at least one” can be used interchangeably herein. It is also to be noted that the terms “comprising”, “including”, and “having” can be used interchangeably.

[0024] FIG. 1 illustrates a forward perspective of a free-rotating grip **100** according to an

embodiment of the present invention.

[0025] The grip **100** comprises a rigid or semi-rigid interrupted sleeve **102**. The grip **100** is tubular and dimensioned to slide on a barbell. The grip comprises an exterior surface **104**, an inner surface **106**, and a forward surface **108** which may be equivalent to a thickness of the sleeve **102**. The thickness may vary from 0.1 mm thick to 10 mm thick.

[0026] The sleeve **102** may be manufactured from polymeric materials, metal, metal alloys, carbon fiber, or organic materials such as wood or leather. The sleeve **102** is adapted to flex and/or compress slightly when a gripping force is applied to the sleeve **102**.

[0027] In various embodiments, the sleeve **102** comprises a flexible, polymeric sock. On other embodiments, the sleeve **102** comprises tubing such as PVC piping. The flexibility of the grip may be adapted to allow a user to slip the grip **100** onto the barbell with an amount of force equal to or less than the force an average user may apply his hands.

[0028] The grip **100** may vary from 4-10 inches in length, or 5-8 inches in some embodiments, and 1-2 inches in diameter in still further embodiments.

[0029] FIG. **2** illustrates a forward perspective of a free-rotating grip **200** according to an embodiment of the present invention;

[0030] The sleeve **102** does not form an uninterrupted fit around the circumference of the barbell, but rather is interrupted by a recess **202** (i.e., slit **202**) which serves the mutual function of providing means to snap the grip **200** onto the barbell and allowing the sleeve **202** to compress in diameter when gripped firmly. The recess **202** may vary from 0.1 mm in width to 50 mm.

[0031] FIG. **3** illustrates an environmental perspective of free-rotating grips **100a-b** according to an embodiment of the present invention.

[0032] The grips **100a-b** are positioned on a barbell **304**. In various embodiments, the grips are spaced apart from one another by a predetermined distance **302**. The distance **302** may change during the bench press exercise as the grips **100a-b** slide laterally to and fro on the barbell **304**. The grips **100a-b** may also rotate axially on the barbell **304**. A user may increase the grip force applied to the grips **100a-b** to force the grips **100a-b** into contact with the barbell **304**, creating a friction fit between the grips **100a-b** and the barbell **304**, increasing the friction between them **304**, **100a-b**.

[0033] FIG. **4** illustrates a forward perspective of a free-rotating grip **400** according to an embodiment of the present invention.

[0034] In various embodiments, the interior surface **106** is affixed to one or more rollers **402a-b** are operable to allow permit axial rotation of the grip **400** about the barbell while increasing resistant to lateral movement of the grip **400** on the barbell. This embodiment may be used for wrist-strengthening exercises. The rollers **402a-b** are non-spherical bearings operable to roll in one direction.

[0035] FIG. **5** illustrates a forward perspective of a free-rotating grip **500** according to an embodiment of the present invention.

[0036] In the shown embodiment, the interior surface **106** comprises a plurality of ball bearings **502a-c** operable to decrease friction between the grip **500** and the barbell **304**, forcing a user to exercise more control and activate more stabilizing muscles to control the barbell **304**.

[0037] FIG. **6** illustrates a forward perspective of a free-rotating grip **600** according to an embodiment of the present invention.

[0038] In some embodiments of the present invention, the interior surface **106** is textured with a repeating or non-repeating pattern **602** to increase friction between the sleeve **600** and the barbell **304**. The texturing **602** may comprise engraved and/or relief texturing. The texturing **602** may also be affixed, or incorporated into, the exterior surface **104** to increase friction with the hands of a user.

[0039] FIG. **7** illustrates a forward perspective of a free-rotating grip according to an embodiment of the present invention.

[0040] In various embodiments, the recess **202** spans 1% to 179% of the circumferences of the

sleeve **700**. The recess **202**, in some embodiments, may span 3% to 125% of the circumference of the sleeve **700**.

[0041] The sleeve **700** defines a hollow interior recess **702** adapted to receive the barbell **304**. The sleeve **700** detachably affixes, or snaps in some embodiments, over the barbell **304**.

[0042] FIG. **8** illustrates a forward perspective of a free-rotating grip and pry bar assembly **800** according to an embodiment of the present invention.

[0043] The grip assembly **800** includes a pry bar **802**. The pry bar **802** may comprise an elongated, oblong, ovoid, or rectangular implement adapted for handheld use. The pry bar **802** may comprise polymeric or metal alloy materials and is operable to pry up a corner of the grip **700** releases said grip **700** from a barbell.

[0044] In various embodiments, the sleeve/grip body **102** is bifurcated into two halves **804a-b** along at a seam **806**, which may be along a top surface. The two halves **804a-b** may be hingedly affixed. The body **102** may clamp down on the barbell **304** and be held in place using clamps, buckles, or other means known to those of skill in the art.

[0045] The present invention may be embodied in other specific forms without departing from its spirit or essential characteristics. The described embodiments are to be considered in all respects only as illustrative and not restrictive. The scope of the invention is, therefore, indicated by the appended claims rather than by the foregoing description. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

Claims

1. A free-rotating, detachable grip for a barbell, the grip comprising: an interrupted, semi-rigid sleeve dimensioned to have an interior circumference exceeding an exterior circumference of a barbell; wherein the sleeve comprises an exterior surface, a forward surface, and an interior surface, the sleeve defining a hollow interior recess adapted to receive a barbell; wherein the sleeve defines a slit operable to allow the grip to detachably affix over a barbell.
 2. The grip of claim 1, wherein the grip comprises polymeric materials.
 3. The grip of claim 1, wherein the interior surface comprises texturing.
 4. The grip of claim 1, wherein the interior surface comprises a plurality of rollers.
 5. The grip of claim 1, wherein the interior surface comprises a plurality of ball bearings.
 6. The grip of claim 1, wherein the grip is operable to slide laterally on a barbell when grip pressure is released.
 7. The grip of claim 1, wherein the grip is operable to slide axially on a barbell when grip pressure is released.
 8. A free-rotating, detachable grip for a barbell, the grip comprising: an interrupted, semi-rigid sleeve dimensioned to have an interior circumference exceeding an exterior circumference of a barbell; wherein the sleeve comprises an exterior surface, a forward surface, and an interior surface, the sleeve defining a hollow interior recess adapted to receive a barbell; wherein the sleeve defines a slit operable to allow the grip to detachably affix over a barbell; wherein the grip is operable to slide laterally and axially on a barbell when grip pressure is released; wherein the grip is operable to lock in place forming a friction fit with the barbell when grip pressure is increased.
-